Households in Western North Carolina Without a Vehicle, from a Disaster Context

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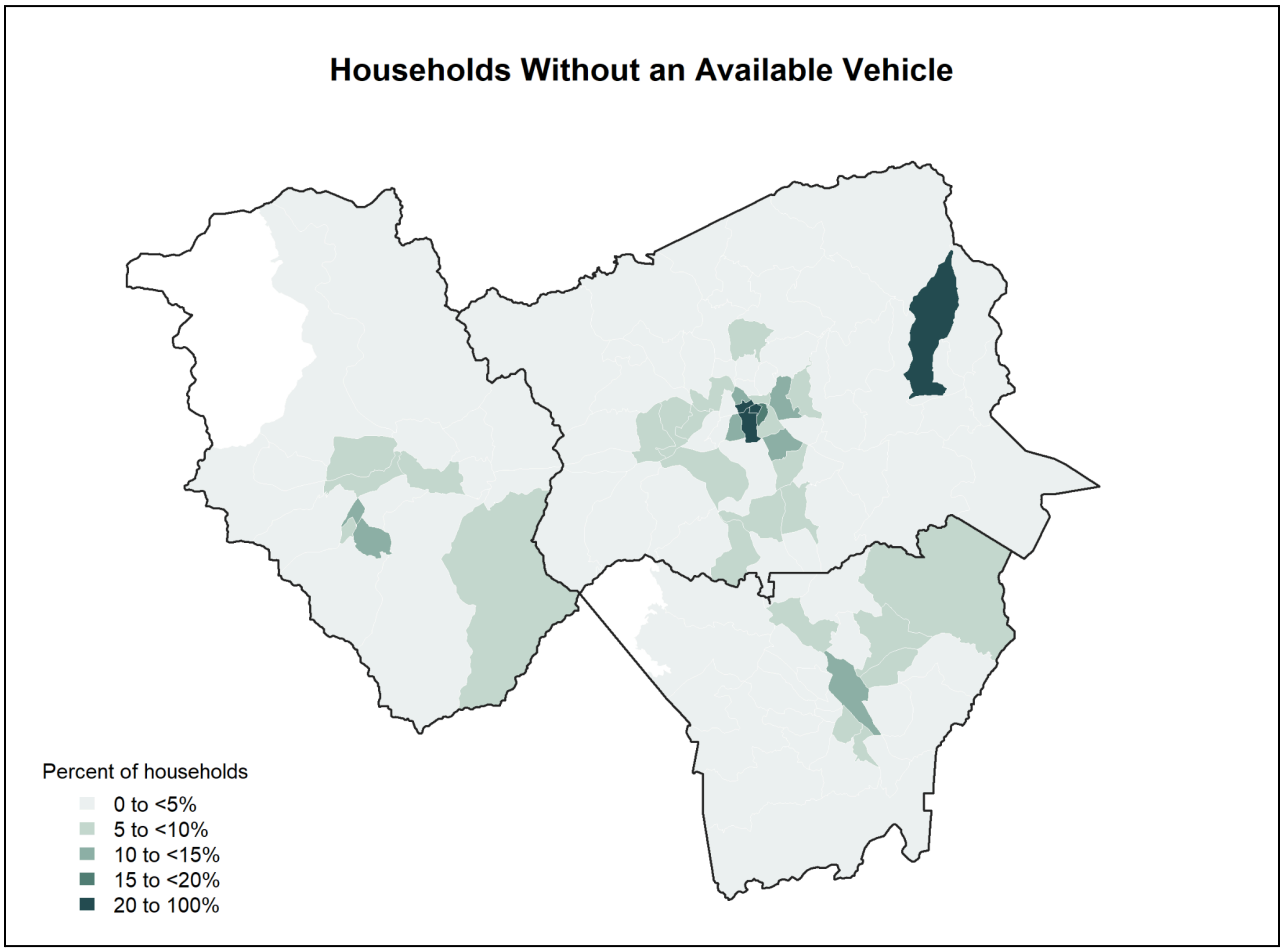
Question and Data

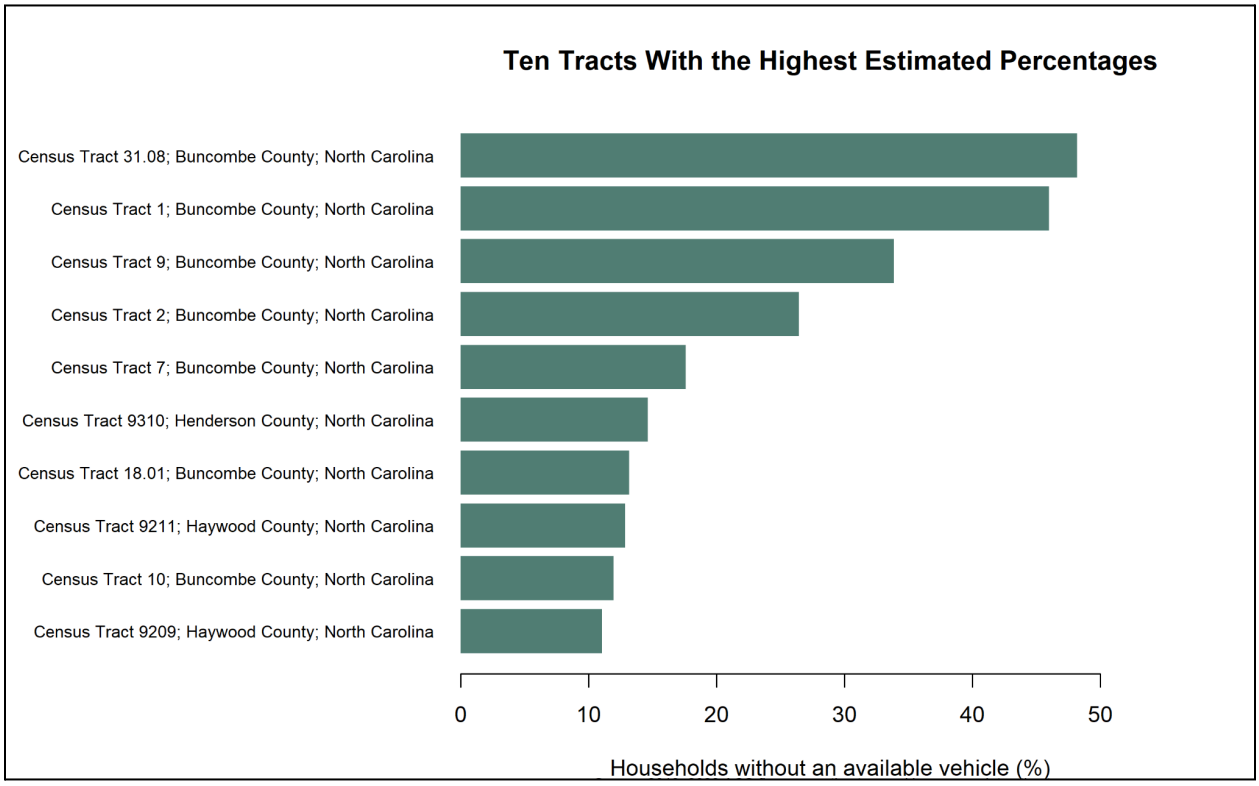
I decided to focus on Western North Carolina in a disaster context, because it is my home state, and Hurricane Helene devastated the area two years ago. Western North Carolina is home to many people in lower economic classes, which is part of why Helene was so devastating. Having access to a car is very important in a possible disaster for evacuation, getting supplies, and reaching assistance.

This project examines geographic differences in household vehicle availability across census tracts in Buncombe, Haywood, and Henderson counties. I used U.S. Census Bureau American Community Survey (ACS) five-year estimates covering 2019-2023, along with the associated census-tract boundaries. Variable B08201_001 represents the total number of households, and B08201_002 represents households with no vehicle available. Using these variables, I calculated and mapped the percentage of households without an available vehicle in each tract.

Major Analysis Steps

First, I downloaded ACS household vehicle data and census-tract boundaries for Buncombe, Haywood, and Henderson counties. Next, I projected the data into the same coordinate system and removed missing or invalid values from the analysis. I created a function to calculate the percentage of households without an available vehicle in each census tract. Finally, I mapped these percentages to compare patterns across the three counties and created a bar chart showing the ten tracts with the highest estimated percentages. I saved the maps and results for my report.





Results

The results showed clear differences in household vehicle availability across the three counties. Most mapped census tracts fell in the lowest category, with fewer than 5% of households lacking an available vehicle. Higher percentages were concentrated in several parts of Buncombe County, with smaller pockets in Haywood and Henderson counties.

Seven of the ten tracts with the highest estimated percentages were in Buncombe County. Census Tract 31.08 had the highest percentage at approximately 48%, followed by Census Tract 1 at approximately 46%. Henderson County's Census Tract 9310 had approximately 15% of households without a vehicle, while Haywood County's highest-ranked tract, Census Tract 9211, had approximately 13%.

These patterns suggest that transportation assistance could be especially valuable in certain communities during a disaster. However, vehicle availability alone does not show whether a household can evacuate safely, since public transportation and support from others may also affect access. These ACS estimates describe conditions before Hurricane Helene, and differences between tracts should be interpreted with their survey margins of error in mind.